

Total No. of Questions : 4]

SEAT No. :

PC-317

[Total No. of Pages : 1

[6361]-189

B.E. (Artificial Intelligence and Data Science) (Insem.)

DATA MODELING & VISUALIZATION

(2019 Pattern) (Semester - VII) (417522)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figure to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain Covariance and Central Limit Theorem. [5]

b) Differentiate between Discrete and Continuous random variables with the help of an example. [5]

c) Define Independent Variable and explain types of Independent Variables with Example. [5]

OR

Q2) a) Explain Data Modelling Concepts with Example. [5]

b) Explain Chebyshev inequality with the help of an example. [5]

c) Define Descriptive Statistics and Graphical Statistics. Explain different Estimation Methods. [5]

Q3) a) Explain Stochastic Processes with the help of an Example. [5]

b) Calculate Pi Using Monte Carlo method. [5]

c) Discuss type 1 and type 2 errors with an example. [5]

OR

Q4) a) Explain the steps of Hypothesis Testing. [5]

b) Differentiate between T-Test , Z-Test and F-Test. [5]

c) Explain Transition State Diagram and Emission State Diagram of Hidden Markov Model with the help of example. [5]

